

ChaTSFM: Channel Adapter for Frozen Time Series Foundation Models (Simplified Reproduction)

Model Research Report (Sample) — based on Li et al., *Channel Adapter for Time Series Foundation Models in Zero-Shot Multivariate Forecasting* (ChaTSFM, ICML 2026)

1. Overview

Time series foundation models (TSFMs) are typically pre-trained channel-independently, so they under-use inter-channel correlations. ChaTSFM is a lightweight plug-and-play adapter (well under 1% extra parameters) that lets a FROZEN backbone exploit multivariate correlations in a zero-shot manner. This simplified reproduction keeps the paper's three pillars: (1) a budgeted synthetic pre-training corpus covering diverse inter-channel dependency patterns; (2) data-derived per-channel 'domain descriptors' that condition an inter-channel similarity measure, reducing cross-domain metric distortion; (3) gated, sparse refinement that blends other channels' backbone forecasts into each channel's prediction without degrading intra-channel temporal dynamics. Implement in pure NumPy/SciPy/scikit-learn.

2. Model Formulation

Frozen backbone (fit once, never updated afterwards): per channel, ridge regression from lag features (lags 1..12, plus sin/cos of time-of-day with period 24) predicting the next $H = 4$ values. Treat this as a stand-in for a frozen channel-independent TSFM.

Domain descriptors: for each channel compute the vector $d_c = [\text{mean}, \text{standard deviation}, \text{trend slope}, \text{lag-1 autocorrelation}, \text{spectral entropy of the periodogram}, \text{dominant-period strength}]$, then z-score each component across the channels of the dataset.

Similarity measure: $\text{sim}(i, j) = \exp(-(d_i - d_j)^T M (d_i - d_j))$ with $M = \text{diag}(m)$, m a non-negative 6-vector learned during adapter pre-training (this is the dataset-conditioned metric).

Gated sparse refinement: for channel i , take the top $k = 2$ most similar other channels; final forecast $y_{\text{hat}}_i = (1 - g_i) * f_i + g_i * \sum_j w_{ij} f_j$, where f denotes the frozen backbone forecast, w_{ij} is the normalized similarity weight, and the gate $g_i = \text{sigmoid}(b + c * \text{mean-similarity-to-top-k})$ with scalars b, c learned. Only (m, b, c) are trainable: 8 parameters total.

3. Hyper-parameters

Horizon $H = 4$; lags 1..12; period 24; top- $k = 2$; ridge $\alpha = 1.0$. Budgeted pre-training corpus: $S = 6$ synthetic multivariate datasets, each with $C = 6$ channels and $T = 600$ timesteps, covering these dependency patterns: lead-lag copies, instantaneous scaling, shared seasonality with channel-specific noise, a regime shift in correlation, and fully independent channels. Optimize (m, b, c) with `scipy.optimize.minimize` (Nelder-Mead, `maxiter = 80`), initializing $m = \text{ones}(6)$, $b = -2$, $c = 4$. All data generation seeded with 42 (pre-training) and 7 (evaluation). Target runtime under 3 minutes on CPU.

4. Training and Evaluation

Zero-shot evaluation: generate 2 fresh synthetic multivariate test datasets (seed 7) with mixed dependency patterns that were never used during adapter pre-training. For each dataset fit a fresh frozen backbone, apply the pre-trained adapter, and compare MSE on the last 20% of the series: backbone-only vs backbone + adapter, per channel and averaged. Print the per-dataset average relative MSE improvement. Success criterion: the adapter improves (or at least matches) average MSE on both zero-shot test datasets, mirroring the paper's consistent gains across nine benchmarks.

5. Implementation Notes

Implement in Python with `numpy`, `scipy`, `scikit-learn` and `pandas` only. Provide classes `FrozenChannelIndependentForecaster` (fit/predict) and `ChaTSFMAdapter` (pretrain(datasets), adapt(X, backbone)) plus a runnable `main.py` demo script: if no CSV file is provided, generate the pre-training and evaluation datasets with the fixed seeds, pre-train the adapter, run the zero-shot evaluation, and print the results.

6. Reference

Dongyuan Li, Renhe Jiang, Shun Zheng, Zheng Dong, Haotian Gao, Ying Zhang, Jiang Bian. Channel Adapter for Time Series Foundation Models in Zero-Shot Multivariate Forecasting. ICML 2026. OpenReview: <https://openreview.net/forum?id=OJriSoFuDq> . Code: <https://github.com/Clearloveyuan/ChaTSFM>